Chapter 14 — Atmosphere

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Weather occurs in:

(a) Thermosphere **(b)** Mesosphere **(c)** Stratosphere **(d)** Troposphere

2. Ozone layer is in:

(a) Mesosphere **(b)** Stratosphere **(c)** Thermosphere **(d)** Troposphere

3. Coldest atmospheric layer:

(a) Troposphere **(b)** Thermosphere **(c)** Stratosphere **(d)** Mesosphere

4. Hottest atmospheric layer:

(a) Thermosphere **(b)** Stratosphere **(c)** Mesosphere **(d)** Troposphere

5. Major gas in atmosphere:

(a) Argon **(b)** Oxygen **(c)** Nitrogen **(d)** CO_2

6. Oxygen percentage about:

(a) 21% **(b)** 78% **(c)** 1% **(d)** 50%

7. Primary pollutant example:

(a) CO **(b)** Ozone **(c)** Acid mist **(d)** PAN

8. Secondary pollutant example:

(a) NO **(b)** Ozone **(c)** SO_2 **(d)** CO

9. CO forms by:

(a) Incomplete combustion **(b)** Photosynthesis **(c)** Respiration only **(d)** Complete combustion

10. NOx forms mainly in:

(a) Trees **(b)** Oceans **(c)** Vehicle engines **(d)** Ice

11. SO_2 comes from:

(a) NaCl only **(b)** Sulfur-containing fuels **(c)** Oxygen only **(d)** Pure water

12. CFCs destroy:

(a) CO_2 **(b)** Stratospheric ozone **(c)** Tropospheric oxygen **(d)** Nitrogen

13. Ground-level ozone is:

(a) Inert **(b)** Harmful pollutant **(c)** Nutrient **(d)** Protective only

14. PAHs are:

(a) Metal ions **(b)** Salts **(c)** Organic pollutants **(d)** Noble gases

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15. POPs are:

(a) Primary oxide particles **(b)** Inert gases **(c)** Persistent organic pollutants **(d)** Pure oxygen pollutants

16. PM affects mainly:

(a) Bones only **(b)** Teeth only **(c)** Hair only **(d)** Lungs

17. CO is toxic because:

(a) Binds haemoglobin **(b)** Destroys ozone **(c)** Burns skin **(d)** Forms acid rain

18. Acid rain contains:

(a) H_2SO_4 and HNO_3 **(b)** NaOH and KOH **(c)** CH_4 only **(d)** NaCl only

19. Acid rain damages:

(a) Only glass **(b)** Only wood **(c)** Marble buildings **(d)** Only plastics

20. Greenhouse gas example:

(a) O_2 **(b)** CO_2 **(c)** N_2 **(d)** Ar

21. Greenhouse gases absorb:

(a) Only UV **(b)** Sound **(c)** Only visible **(d)** Infrared radiation

22. Deforestation increases:

(a) Atmospheric CO_2 **(b)** Nitrogen only **(c)** Oxygen only **(d)** Ozone only

23. Photochemical smog contains:

(a) Only smoke **(b)** Only CO_2 **(c)** Ozone and PAN **(d)** Only SO_2

24. Brown colour in smog due to:

(a) NO_2 **(b)** CO **(c)** O_2 **(d)** CO_2

25. Catalytic converters reduce:

(a) Only dust **(b)** Only noise **(c)** Vehicle pollutants **(d)** Only CO_2

26. Renewable energy reduces:

(a) Only water use **(b)** Only soil pH **(c)** Only noise **(d)** Air pollution

27. Volcanoes are:

(a) Only ozone source **(b)** Natural pollution source **(c)** Human source only **(d)** No source

28. VOCs are:

(a) Noble gases **(b)** Volatile organic compounds **(c)** Metal vapours **(d)** Very oxidized carbons

29. Lead pollution affects:

(a) Only skin **(b)** Nervous system **(c)** Only hair **(d)** Only bones

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30. Enhanced greenhouse effect causes:

(a) Acid rain only **(b)** Global warming **(c)** Smog only **(d)** Ozone hole only

31. Weather occurs in:

(a) Troposphere **(b)** Mesosphere **(c)** Stratosphere **(d)** Thermosphere

32. Ozone layer is in:

(a) Mesosphere **(b)** Troposphere **(c)** Stratosphere **(d)** Thermosphere

33. Coldest atmospheric layer is:

(a) Troposphere **(b)** Thermosphere **(c)** Mesosphere **(d)** Stratosphere

34. Primary pollutant example:

(a) Acid mist **(b)** Ozone **(c)** CO **(d)** PAN

35. Secondary pollutant example:

(a) Ozone **(b)** CO **(c)** NO **(d)** SO_2

36. CO forms by:

(a) Incomplete combustion **(b)** Respiration only **(c)** Photosynthesis **(d)** Complete combustion

37. CO_2 is a:

(a) Primary toxic metal **(b)** Greenhouse gas **(c)** Noble gas **(d)** Halogen

38. NOx contributes to:

(a) Only greenhouse effect **(b)** Smog and acid rain **(c)** Only ozone layer **(d)** No pollution

39. SO_2 contributes to:

(a) Metal corrosion only **(b)** Acid rain **(c)** No pollution **(d)** Ozone formation only

40. CFCs destroy:

(a) Nitrogen **(b)** Stratospheric ozone **(c)** Tropospheric oxygen **(d)** CO_2

41. Ground-level ozone is:

(a) Noble gas **(b)** Harmful oxidant **(c)** Metal **(d)** Protective UV shield

42. PAHs are:

(a) Metal oxides **(b)** Primary acid halides **(c)** Polycyclic aromatic hydrocarbons **(d)** Pure air hydrocarbons

43. POPs are:

(a) Pure organic particles **(b)** Persistent organic pollutants **(c)** Primary oxygen pollutants **(d)** Metal salts

44. VOCs are:

(a) Metal vapours **(b)** Very oxidized carbons **(c)** Noble gases **(d)** Volatile organic compounds

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45. PM2.5 is dangerous because:

(a) Is always visible **(b)** Is a gas **(c)** Penetrates deep lungs **(d)** Is harmless

46. Acid rain contains:

(a) NaCl only **(b)** H_2SO_4 and HNO_3 **(c)** CH_4 only **(d)** NaOH and KOH

47. Greenhouse gas example:

(a) O_2 **(b)** Ar **(c)** CO_2 **(d)** N_2

48. Photochemical smog needs:

(a) Sunlight, NOx and VOCs **(b)** Only SO_2 **(c)** Only CO_2 **(d)** Darkness only

49. AQI summarizes:

(a) Soil pH **(b)** Noise only **(c)** Air pollutant levels **(d)** Water purity

50. Nephelometer measures:

(a) CO_2 only **(b)** Temperature only **(c)** pH **(d)** Light scattering by particles

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Answer Key with Brief Reasons

1. **Answer: (d) Troposphere** — Lowest layer.
2. **Answer: (b) Stratosphere** — Protective layer.
3. **Answer: (d) Mesosphere** — Temperature minimum.
4. **Answer: (a) Thermosphere** — High-energy absorption.
5. **Answer: (c) Nitrogen** — About 78%.
6. **Answer: (a) 21%** — Book value.
7. **Answer: (a) CO** — Emitted directly.
8. **Answer: (b) Ozone** — Formed in air.
9. **Answer: (a) Incomplete combustion** — Limited oxygen.
10. **Answer: (c) Vehicle engines** — High-temperature combustion.
11. **Answer: (b) Sulfur-containing fuels** — Combustion.
12. **Answer: (b) Stratospheric ozone** — Chlorine radicals.
13. **Answer: (b) Harmful pollutant** — Photochemical smog.
14. **Answer: (c) Organic pollutants** — Polycyclic aromatic hydrocarbons.
15. **Answer: (c) Persistent organic pollutants** — Resist degradation.
16. **Answer: (d) Lungs** — Particulate matter.
17. **Answer: (a) Binds haemoglobin** — Reduces O_2 transport.
18. **Answer: (a) H_2SO_4 and HNO_3** — From SOx/NOx.
19. **Answer: (c) Marble buildings** — Reacts with carbonates.
20. **Answer: (b) CO_2** — Absorbs IR.
21. **Answer: (d) Infrared radiation** — Heat trapping.
22. **Answer: (a) Atmospheric CO_2** — Less uptake.
23. **Answer: (c) Ozone and PAN** — Sunlight reactions.
24. **Answer: (a) NO_2** — Brown gas.
25. **Answer: (c) Vehicle pollutants** — CO/NOx/HC.
26. **Answer: (d) Air pollution** — Cleaner source.
27. **Answer: (b) Natural pollution source** — Ash/gases.
28. **Answer: (b) Volatile organic compounds** — Organic pollutants.
29. **Answer: (b) Nervous system** — Heavy metal toxicity.
30. **Answer: (b) Global warming** — Climate change.
31. **Answer: (a) Troposphere** — Lowest layer.

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32. Answer: (c) Stratosphere — UV absorption.

33. Answer: (c) Mesosphere — Mesopause.

34. Answer: (c) CO — Emitted directly.

35. Answer: (a) Ozone — Formed in atmosphere.

36. Answer: (a) Incomplete combustion — Limited oxygen.

37. Answer: (b) Greenhouse gas — IR absorption.

38. Answer: (b) Smog and acid rain — NO/NO_2.

39. Answer: (b) Acid rain — H_2SO_4 formation.

40. Answer: (b) Stratospheric ozone — Cl radicals.

41. Answer: (b) Harmful oxidant — Smog component.

42. Answer: (c) Polycyclic aromatic hydrocarbons — Incomplete combustion.

43. Answer: (b) Persistent organic pollutants — Resist degradation.

44. Answer: (d) Volatile organic compounds — Evaporate easily.

45. Answer: (c) Penetrates deep lungs — Fine particles.

46. Answer: (b) H_2SO_4 and HNO_3 — Acidic deposition.

47. Answer: (c) CO_2 — IR absorption.

48. Answer: (a) Sunlight, NOx and VOCs — Secondary pollutants.

49. Answer: (c) Air pollutant levels — Air quality index.

50. Answer: (d) Light scattering by particles — Particulate monitoring.

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